

Experiment 4

Practical Exercise: Build and Run a Java Application with Maven, Migrate the Same Application to Gradle

Part 1: Create and Build a Java Application with Maven

Step 1: Create a Maven Project in IntelliJ IDEA

1. Open IntelliJ IDEA

Launch **IntelliJ IDEA** and click on **File** → **New** → **Project**.

2. Select Maven

In the **New Project** window, choose **Maven** from the options on the left.

Check **Create from archetype** and select **maven-archetype-quickstart**.

Click **Next**.

3. Enter Project Details

GroupId: com.example

ArtifactId: MVNGRDLDEMO

Click **Next** and then **Finish**.

4. Wait for IntelliJ to Load Dependencies

IntelliJ will automatically download the Maven dependencies, so just relax for a moment.

Step 2: Update pom.xml to Add Build Plugin

To compile and package your project into a .jar file, you need to add the **Maven Compiler** and **Jar** plugins. 

a) 1. Open the pom.xml file. 

2. Add the following inside the <project> tag:

```
<build>
<plugins>
<!-- Compiler Plugin -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
<!-- Jar Plugin -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-jar-plugin</artifactId>
<version>3.2.0</version>
<configuration>
```

```
<archive>
<manifest>
<addClasspath>true</addClasspath>
<mainClass>org.example.Main</mainClass>
</manifest>
</archive>
</configuration>
</plugin>
</plugins>
</build>
```

b) 1. Open the Main.java file. 

2. Add the following code

```
package org.example;

public class Main {
    static void main() {
        System.out.println("Hello students");
        System.out.println("Welcome to Devops Lab");
        System.out.println("Lab Program no.4");
    }
}
```

Step 3: Build and Run the Maven Project

1. Open IntelliJ IDEA Terminal

Press **Alt + F12** to open the terminal. 

2. Compile and Package the Project

Run the following commands to build the project:

```
mvn clean compile
mvn package
```

3. Locate the JAR File

After running the above, your .jar file will be located at:

```
D:\Idea Projects\MVNGRDLDEMO\target\MVNGRDLDEMO-1.0-SNAPSHOT.jar
```

4. Run the JAR File

To run the generated JAR file, use:

```
java -jar target\MVNGRDLDEMO-1.0-SNAPSHOT.jar
```

Part 2: Migrate Maven Project to Gradle

Step 1: Initialize Gradle in Your Project

1. Open Terminal in IntelliJ IDEA

Make sure you're in the project directory:

```
cd "D:\Idea Projects\MVNGRDLDEMO"
```

2. Run Gradle Init Command

Execute the following command to migrate your Maven project to Gradle:

```
gradle init --type pom
```

This command will convert your Maven pom.xml into a Gradle build.gradle file.

Step 2: Review and Update build.gradle

1. **Open** build.gradle in IntelliJ IDEA.
2. Ensure the following configurations are correct:

```
plugins {  
    id 'java'  
}  
group = 'com.example'  
version = '1.0-SNAPSHOT'  
repositories {  
    mavenCentral()  
}  
dependencies {  
    testImplementation 'junit:junit:4.13.2'  
}  
jar {  
    manifest {  
        attributes('Main-Class': 'org.example.Main')  
    }  
}
```

Step 3: Build and Run the Gradle Project

1. Clean and Build the Project

To clean and build your Gradle project, run:

```
gradle clean build
```

2. Run the Generated JAR File

Now, run the generated JAR file using:

```
java -jar build/libs/MVNGRDLDEMO-1.0-SNAPSHOT.jar
```